



**STUDIES ON SOME PROPERTIES OF SNAIL SHELL AND CASSAVA PEEL
POWDERED POLYPROPYLENE COMPOSITES.**

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ABSTRACT

This study investigated some properties of snail and cassava peel-filled polypropylene composites. The snail shell and cassava peel powders of particle sizes ($90\mu\text{m}$) were incorporated into the polymer polypropylene matrix in a ratio of 14:3:3, 15:3:2 and 16:3:1 for polypropylene/snail shell/cassava peel; 17:3, 18:2 and 19:1 for polypropylene/cassava peel and 17:3, 18:2, 19:1 for polypropylene/snail shell powdered composites. The mixtures were prepared using injection molding machines at a hopper temperature of 200°C and at a rotor speed of 60rpm and at an average thickness of 3.00mm in accordance with American Society of Testing and Materials Standard, to determine the mechanical properties of the composite. The rates of bio-degradation and water absorption of the composites were also investigated for 120 days and 42 days respectively. Results showed that the tensile properties, the flexural properties as well as impact strength of the composite decreased with reference to the same properties of the neat polypropylene. However, the tensile properties, flexural properties and impact strength to these composites increased with increasing filler content loading. Again, from the results, the rate of bio-degradation and water absorption increased with increasing filler content loading. The results showed that the incorporation of cassava peel and snail shell powders into polypropylene matrix decreased the mechanical properties of the composite while the rate of bio-degradation and water absorption increased with time.

Key words: Polypropylene, Biodegradation, Snail, Cassava, Composite, Absorption

1.0 Biodegradation of organic materials by microorganisms like fungi and bacteria, is defined as the process by which this happens (Vart et al, 2012). Biodegradable plastics are plastics that can be broken down into low-weight chemicals and non-toxic byproducts once they have served their useful purpose (Vander., 2011). There are three stages to the biodegradation process; the first of which is bio-deterioration, which occurs when a material is exposed to the abiotic factors of the outdoor environment and undergoes a series of undesirable actions that weaken its mechanical, physical, and chemical properties. Temperature, light, oxygen, water, and chemicals are all examples of abiotic variables (Muller, 2005; Hueck, . 1996). The second phase of biodegradation, known as bio-fragmentation, is a lytic process in which a bond inside a polymer is severed, resulting in the generation of oligomers and monomers. Both aerobic and anaerobic digestion could be used to break down the material (Lucas, . *et al.*, 2008). Assimilation is the third phase of biodegradation (Lucas, . *et al.*, 2008), and it involves the incorporation of the bio-fragmentation by-product into the microbial cells.

Compositing stemmed from three main reasons, which include reduction in cost of production, strength improvement and enhancement of biodegradability (Carraher, 2014). In the reduction of cost in production, it is observed that the materials used as fillers are mostly organic waste and such are acquired at little or no cost compared to ceramic materials and other inorganic materials. Plastic materials are more costly than these supposedly worthless organic wastes. Thus, compositing reduces the amount of plastic materials used thereby reducing the cost of production of these materials (Carraher, 2014; Shama *et al*, 2018). For strength improvement, these composite materials contain non-plastic substances that constitute mostly of organic fibers, these non-plastic materials tend to act as

an interfacial adhesive and have interfacial interaction with plastic constituents (kutnar, and Muthu, 2016). The interfacial adhesion holds plastic particles together. Comparison of the physical characteristics of the composite to those of the pure plastic is an efficient method of confirming this idea (Hueck, 1996).

2.0. EXPERIMENTAL

2.1 Sample Collection and Preparation

Cassava Peel.

The cassava peels were obtained from a local farm near Ovogo in Emohua Local Government Area of Rivers State of Nigeria and it was identified by the curator in the Department of Plant Science and Biotechnology of University of Port Harcourt. The cassava peels were carefully selected to ensure that only the peels were used. The selected cassava peels were thoroughly washed and dried in the sun for 96 hours, and thereafter it was oven dried at one hundred and thirty (130 °C) for 4 hours. The dried cassava peels were ground using a locally fabricated grinding machine and the powder obtained thereafter was sieved using 90 µm sieve to obtain a uniform particle size. The sieved cassava peel powder was oven dried for 4 hours at 130 °C and it was kept in an airtight container for further analysis (Carraher, 2014)

Snail Shell

The snails were sourced from a local market in Choba near the University of Port Harcourt. The shells were carefully removed and thoroughly washed using distilled water. The snail shells were sun dried for 120 hours and thereafter oven dried at 130 °C for five hours. The dried snail shells were ground using locally fabricated grinding machine and the powder obtained thereafter was sieved using 90 µm standard laboratory test sieve to obtain a uniform particle size. The sieved snail shell powder was oven dried at 130 °C for four hours and thereafter, it was stored in an air tight container in order to prevent it from absorbing atmospheric moisture before further analysis.

Polypropylene

The polypropylene (PP) granules (Density 0.956 g/cm³) were obtained from Indorama Petrochemicals, Eleme near Port Harcourt Rivers State, Nigeria.

The pulverized cassava peel powder, snail shell powder and polypropylene granules were mixed in different proportion to produce polypropylene composite as shown in Table 2.1:

2.2 Determination of Moisture Content.

The moisture content of the fillers Snail Shell Powder (SSP) and Cassava Peel Powder (CPP) was determined according to the method as reported by ASTM D570. Each filler (SSP and CPP) was weighed and the initial mass recorded as W_1 . Thereafter, the fillers (SSP and CPP) were oven dried at the temperature of 100 °C until a constant mass was obtained. The final weight of the filler was recorded as W_2 and the percentage of moisture content was calculated using the formula (equation 1):

$$\% \text{ Mc} = \frac{W_1 - W_2}{W_1} \quad \text{Equation 1}$$

Where W_1 = initial weight

W_2 = final weight

Mc = moisture content.

2.3 pH Measurement

The pH of the powder was determined according to the standard procedure for determination of pH of solids. Snail shell powder and cassava peels powder measuring 3g each were dissolved in separate beakers containing 100ml of distilled water. The solution was agitated with a magnetic stirrer for 30 minutes for proper dissolution in water and a pH meter equipped with temperature sensor was used to monitor changes in reading.

2.4 Preparation of Polypropylene Composites

The composites were prepared using an injection molding machine at Ceeplast Industries Limited, Aba in Abia State. The mixing was done in a beaker and later transferred to a hopper of the injection moulding machine at the temperature between 180 °C- 210 °C through the nozzle at the temperature of 235 °C - 285 °C at the rotor speed of 60rpm. The average thickness of the formation was 3.00mm.

Table 2.1: Sample Composition

Samples		Percent weight composition		
PP (g)		SSP(g)		CPP(g)
PURE PP	200	-		-
PP/SSP-10	190	10		-
PP/SSP-20	180	20		-
PP/SSP-30	170	30		-
PP/CPP-10	190	-		10
PP/CPP-20	180	-		20
PP/CPP	170	-		30
PP/SSP-30/CPP-10	160	30		10
PP/SSP-30/CPP-20	150	30		20
PP/SSP-30/CPP-30	140	30		30

PP; polypropylene, **SSP;** Snail Shell Powder, **CPP;** Cassava Peel Powder.

2.4 General Method for Preparation of Polypropylene and Composites.

Pure samples of polypropylene granules measuring 200g were introduced into the hopper of injection molding machine whose temperature was set at 230 °C, the granules were allowed to melt in order to form a uniform paste. However, under the influence of pressure in the hopper, the paste was moved to the nozzle of the injection molding machine and the temperature of the machine was set at 260 °C and the paste obtained was transferred to a mold where it solidified and took the shape of the internal dimension of the mold to cool, collected, labeled appropriately and was stored for further analysis.

The procedure was carried out on the samples of polypropylene granules, snail shell powder and cassava peel powder of different proportion as indicated in Table 3.1. However, in the formulation of composites, the hopper and nozzle temperature are set at 200° C and 230° C respectively.

2.5 Water Absorption

Following the standard procedure for water absorption test, samples of pure polypropylene (PP) materials (plastic) and samples of composite materials of different formulation were cut into six pieces each, at Science and Engineering Workshop, University of Port Harcourt. The dimension of specimens is 25mm x 25mm of 3.00mm thickness. The samples were oven dried at 50 °C for four hours in order to remove absorbed moisture. Each sample was weighed using electric weighing balance, the thickness was also measured using micrometer screw gauge, and the initial weight and the initial thickness were recorded as W_i and L_i respectively. Thereafter, the sample of pure polypropylene (plastic) and the sample of the composites materials of different filler formulation were immersed in a different 400ml of beaker containing 200ml each of distilled water for forty-two days. After an interval of seven days, one sample for each formulation was taken out of the water, dried with towel and then weighed immediately and the final weight was

recorded as (W_f) and the final thickness was also determine and recorded as L_f . The weight gain of each sample was computed as percentage weight (% M_w). The percentage water absorption was calculated by using the equation:

$$M_w (\%) = \frac{W_f - W_i}{W_f} \times 100$$

equation 2

Where M_w (%) =Weight gain

W_f = Final weight

W_i = Initial weight

2.6 Biodegradability Test

Following the standard procedure for evaluating and determining the biodegradation properties of neat PP (plastic) sample and composite materials. Each sample formulation was cut into four pieces that has the dimension of 25mm x 25mm of 3.00mm thickness, these were oven dried at 50 °C for four hours in order to remove absorbed moisture. Each of the sample was weighed and the weight was recorded as W_i . The samples were buried in the ground at the depth of 25cm beneath the ground on 25th June, 2022 for a period of 120 days. Thereafter, one sample of each formulation was exhumed after interval of 30 days. It was thoroughly washed in order to remove soil particles, sun dried for 12 hours and then weighed with electric weighing balance and its final weight was recorded as W_f and the final thickness as L_f . The percentage of reduction in weight of the sample was calculated using the equation:

$$Br (\%) = \frac{W_i - W_f}{W_i} \times 100$$

equation 3

Where W_i = initial weight

W_f = final weight

Br = % of biodegradation

2.7 Mechanical Properties Testing

The sample of neat polypropylene (plastic) and the composite materials of different filler formulation were characterized for some mechanical properties at Engineering Materials Development Institute, Akure in Ondo State.

2.7.1 Tensile Strength

The test for tensile strength was conducted on clean polypropylene and composite components with varying filler formulations. The Universal Instron Checking Machine (model 3369) was used for this purpose, with a constant cross head acceleration rate of 1mm/min. The samples were machined into a dumbbell form with a length of 100mm, a width of 18mm for the two edges, and a width of 13mm for the central part. The samples, each with a thickness of 3.00mm, were then subjected to testing following the standard process outlined in ASTM D 638. The load, duration at the yield point, and the load-extension curve were measured and the provided findings represent the average values for each sample.

2.7.2 Flexural Strength

The flexural strength test of neat polypropylene samples and the composite materials of different filler formulation was determined by using three-point test on the samples. The test was done using Universal testing machine (model 3369) operated at a crosshead speed of 2mm/min. The samples were cut into horizontal bars of length (100mm) and width (18mm) of 3.00mm thickness. The samples were tested one after the other according to the ASTM D790 standard procedure. The load-deflection data was obtained, recorded and graph plotted. The results presented are the average result of each sample.

2.7.3 Impact Strength

The notched impact strength of the neat polypropylene (plastic) samples and the composite materials of different formulation were tested on an impact tester (CG TOYOSEIKI, NO 556 Impact test). Each of the sample was cut into a rectangular bar shape of length (100mm) and the width (18mm) of 3.00mm thickness in accordance with ASTM D6110 standard. Different values were determined and the results presented are the average of each sample.

3.0 RESULTS AND DISCUSSION

The results obtained from the characterization of mechanical properties, rate of water absorption and biodegradation studies on the formulated polypropylene composites are presented in Tables, Figures and graph in chapter three.

3.1 Moisture Content

Moisture content of Cassava Peels Powder (CPP)

Initial weight = 4.01g

Final weight = 3.95g

% weight loss = 1.50 %

Moisture content of snail shells powder (SSP)

Initial weight = 4.00; Final weight = 3.89

% weight loss = 2.75 %

The moisture content of the Cassava Peel Powder (CPP) was found to be 1.5% and that of Snail Shell Powder (SSP) was also found to be 2.75%. It showed that the fillers used for the compositing contains very minimal amount of moisture and so the effects of the moisture content of the fillers were very negligible on the properties of the composites.

3.2 pH Measurement of Cassava Peel (CPP) and Snail Shell Powders (SSP)

CPP = 6.32

SSP = 8.01

The pH of cassava peel and snail shell powders were 6.32 and 8.01 respectively. The pH result showed that SSP is basic which could be attributed to higher content of calcium carbonate while that of CPP showed that it is acidic.

3.3 Biodegradation

The results obtained from the biodegradability studies are presented in tables and graph. Relating to Figures 3.1 to 3.6, the results showed that the rate of biodegradability increased in proportion with its duration in the soil with increase in the amount of filler content loading.

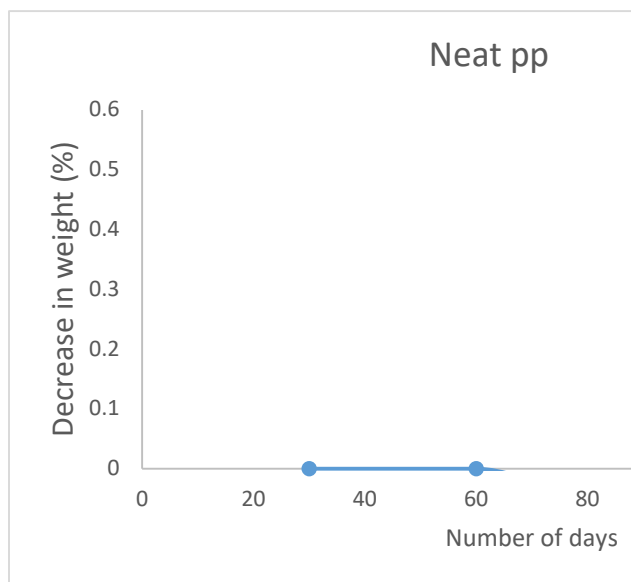


Figure 3.1: Graph of percentage weight loss in biodegradation of neat PP

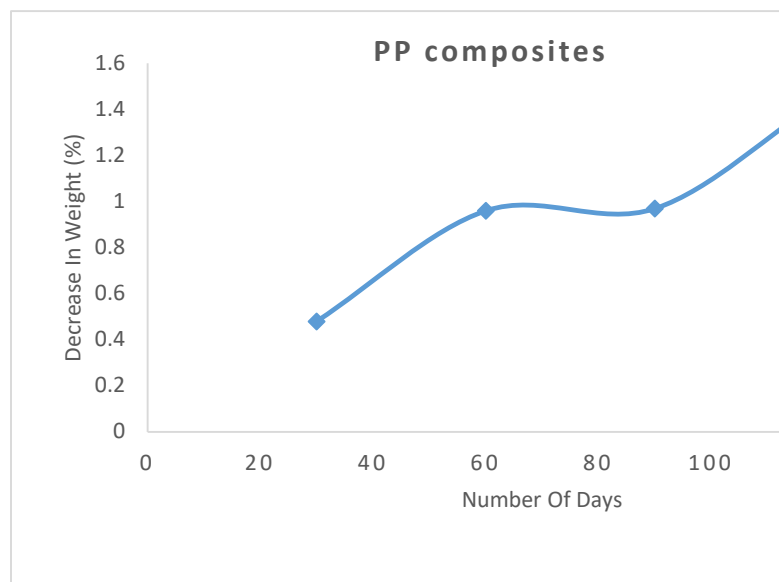


Figure 3.3: Graph showing percentage weight loss in biodegradation of 170g PP/30g SSP

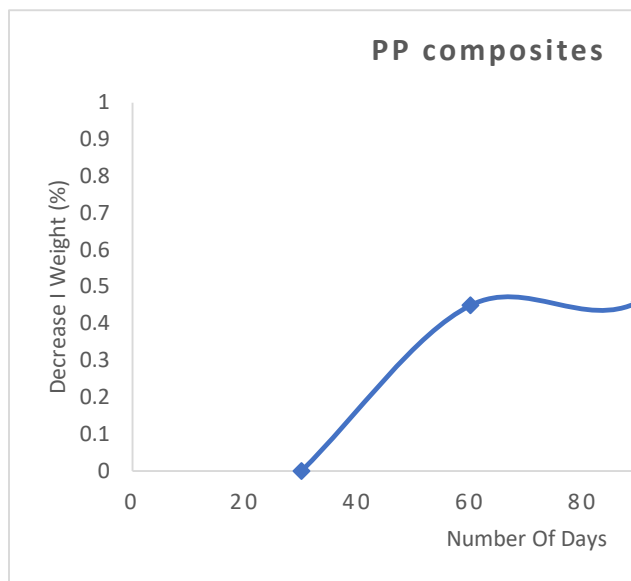


Figure 3.2: Graph showing percentage weight loss in biodegradation of 190g PP/10g SSP

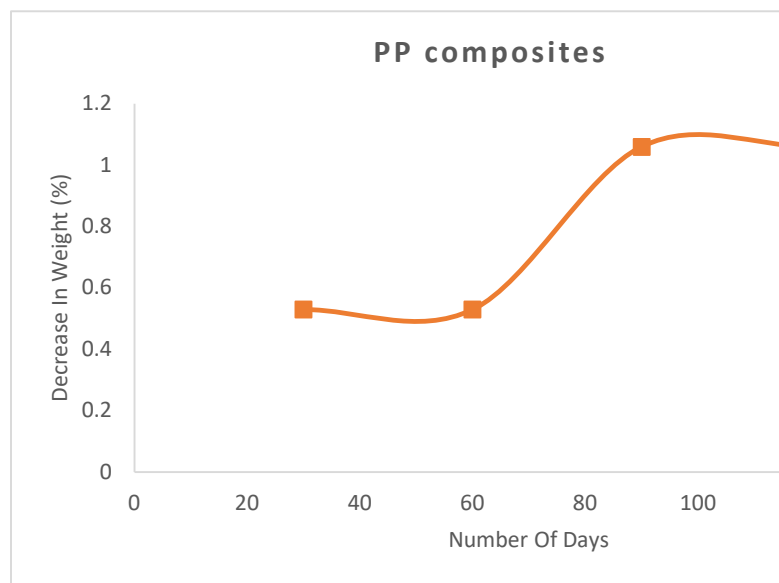


Figure 3.4: Graph showing percentage weight loss in biodegradation of 190g PP/10g CPP

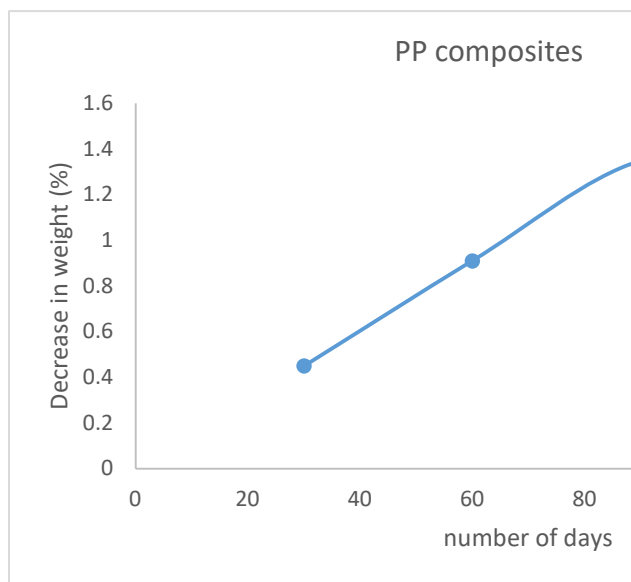


Figure 3.5: Graph showing percentage weight loss in biodegradation of 170g PP/30g CPP

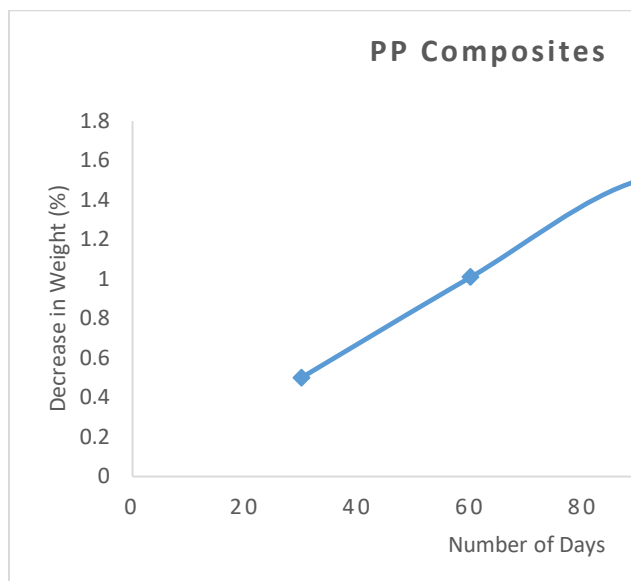


Figure 3.6: Graph showing percentage weight loss in biodegradation of 150g PP/30g SSP/20 CPP

The results showed that the composites responded to the microbial attack after thirty days in the soil. It was also observed that there was continuous microbial attack as long as the composites were buried in the soil. These led to a steady decrease in weight of the composites except the neat PP which maintained a constant weight until after 120 days before a slight decrease in weight was noticed. The results also showed that the rate of biodegradation increased as the amount of filler content of the composites is increased for all the formulation (PP/SSP, PP/CPP, PP/SSP/CPP). This trend was similar to the study by Farahana *et al*, 2015. The rate of biodegradation for composites increased as the filler content loading is increased as a function of time in all formulation.

3.4 Water absorption test

The results from the water absorption studies are presented in table and graph. Referring to figure 3.7 to 3.13 below, the results showed that the rate of water absorption increased with increase in the amount of filler proportional with its duration in water.

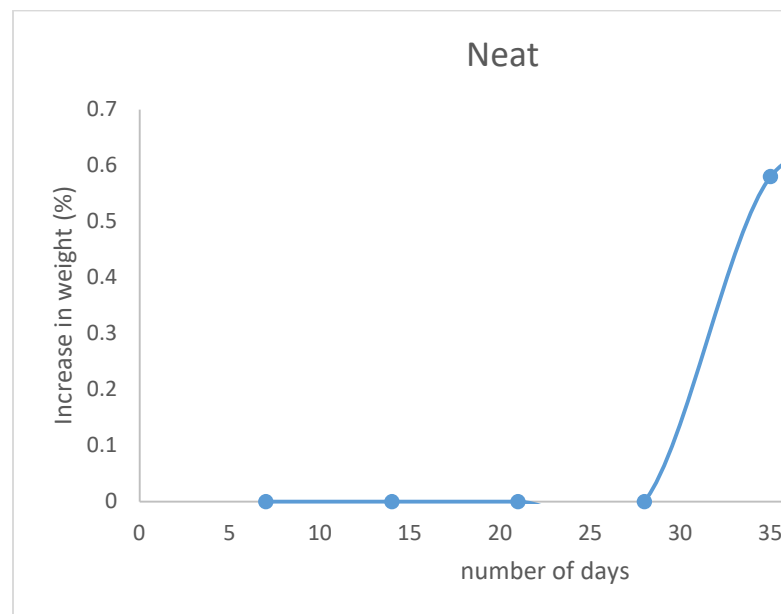


Figure 3.7: Graph showing the percentage weight gain in water absorption neat PP

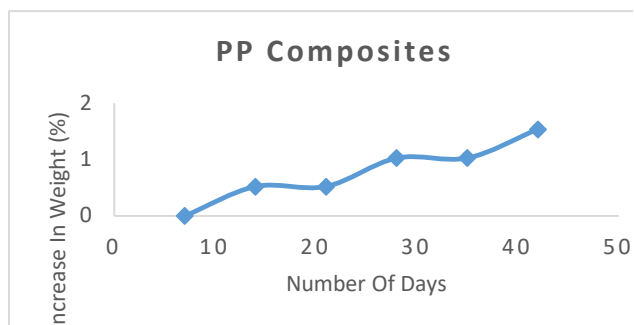


Figure 3.8: Graph showing the percentage weight gain in water absorption of 190g PP/10g SSP

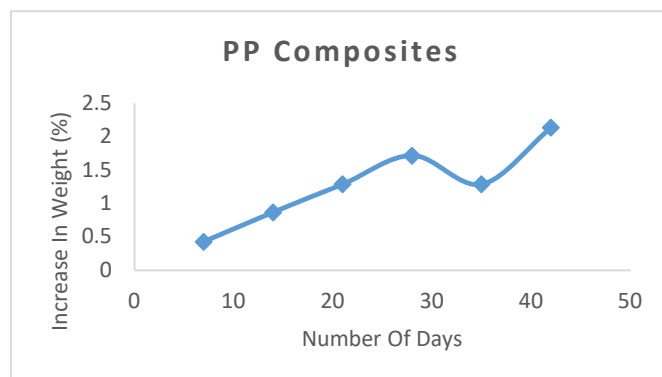


Figure 3.11: Graph showing the percentage weight gain in water absorption of 170g PP/30g CPP

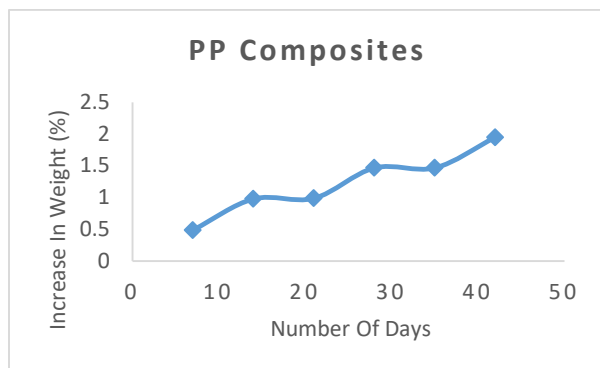


Figure 3.9: Graph showing the percentage weight gain in water absorption of 170g PP/30g SSP.

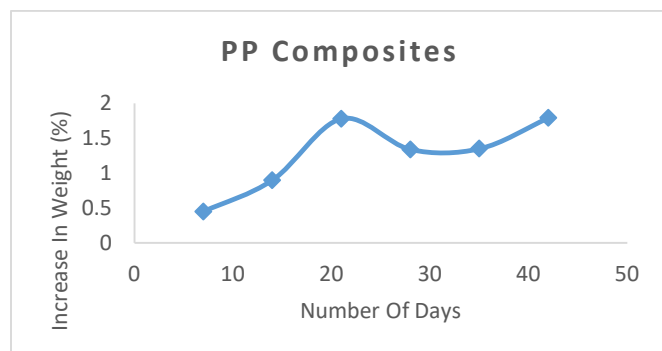


Figure 3.12: Graph showing the percentage weight gain in water absorption of 160g PP/30g SSP/10g CPP

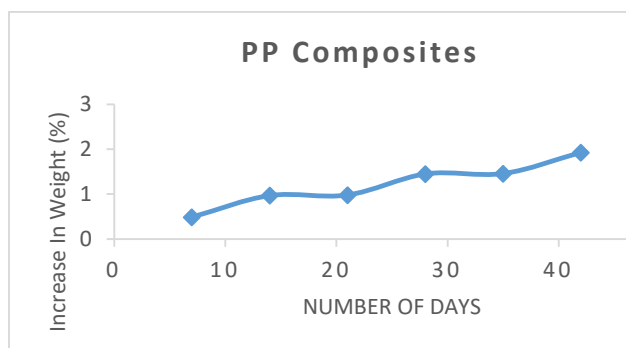


Figure 3.10: Graph showing the percentage weight gain in water absorption of 190g PP/10g CPP

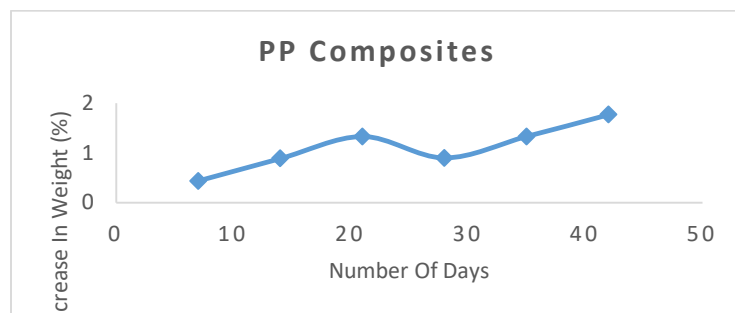


Figure 3.13: Graph showing the percentage weight gain in water absorption of 140g PP/30g SSP/30g CPP.

The water absorption of polypropylene (PP) composites as illustrated in the above figures showed increase in water absorption with increase in filler content in proportion to their duration in water for all the investigated composites. All the prepared polypropylene composites absorbed more water than the neat PP. The filler loadings were incorporated into the polypropylene matrix. It was observed that the rate of water absorption in CPP filled composites was found to be higher than that of SSP and CPP/SSP filled composites. The rate increase of water absorption of CPP filled composite

decreases in the form as $170\text{gPP}/30\text{g CPP} > 180\text{g PP}/20\text{g CPP} > 190\text{g PP}/10\text{g CP}$

3.5 Mechanical Properties.

The results from the mechanical properties studies such as tensile strength, tensile modulus, flexural strength, flexural modulus, elongation at break and impact strength are presented in graph, charts and Tables as shown in Figures 3.14 to 3.17. The results showed that they increased with increase in the amount of filler

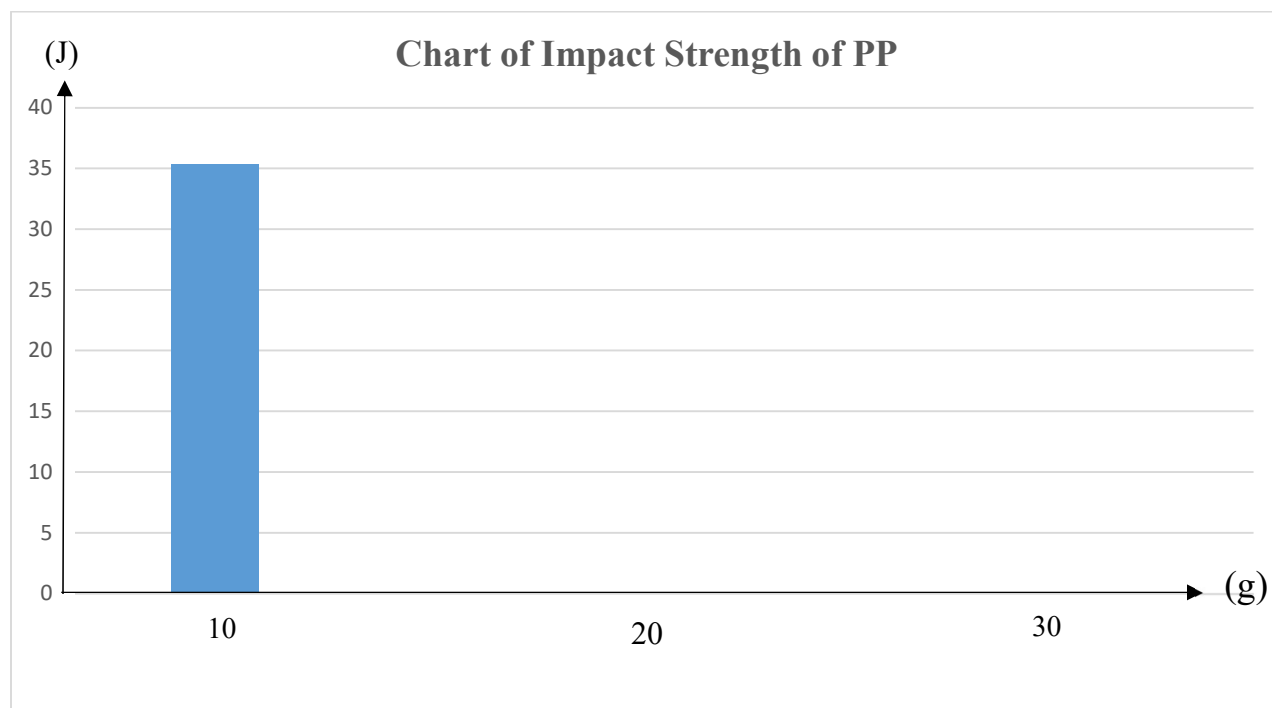


Figure 3.14: Chart Showing Impact Strength of Neat PP

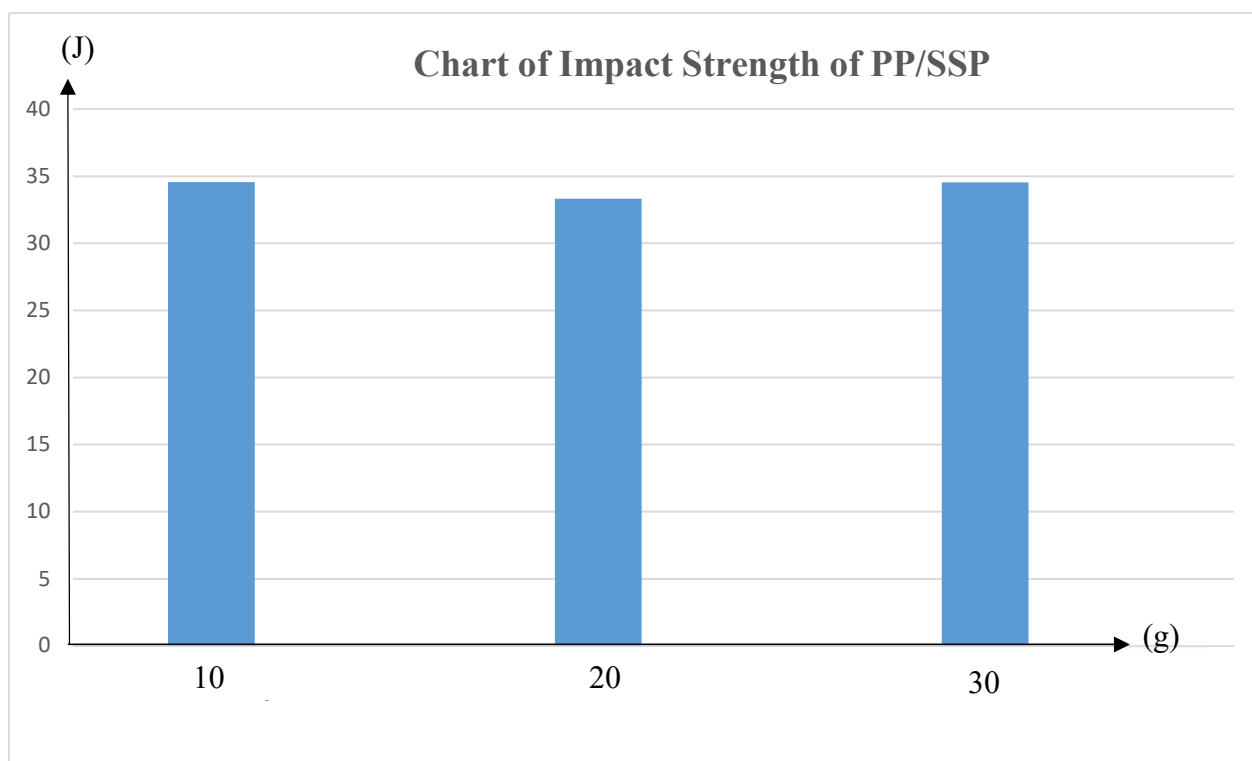


Figure 3.15: Chart Showing Impact Strength of PP/SSP Filled Composites

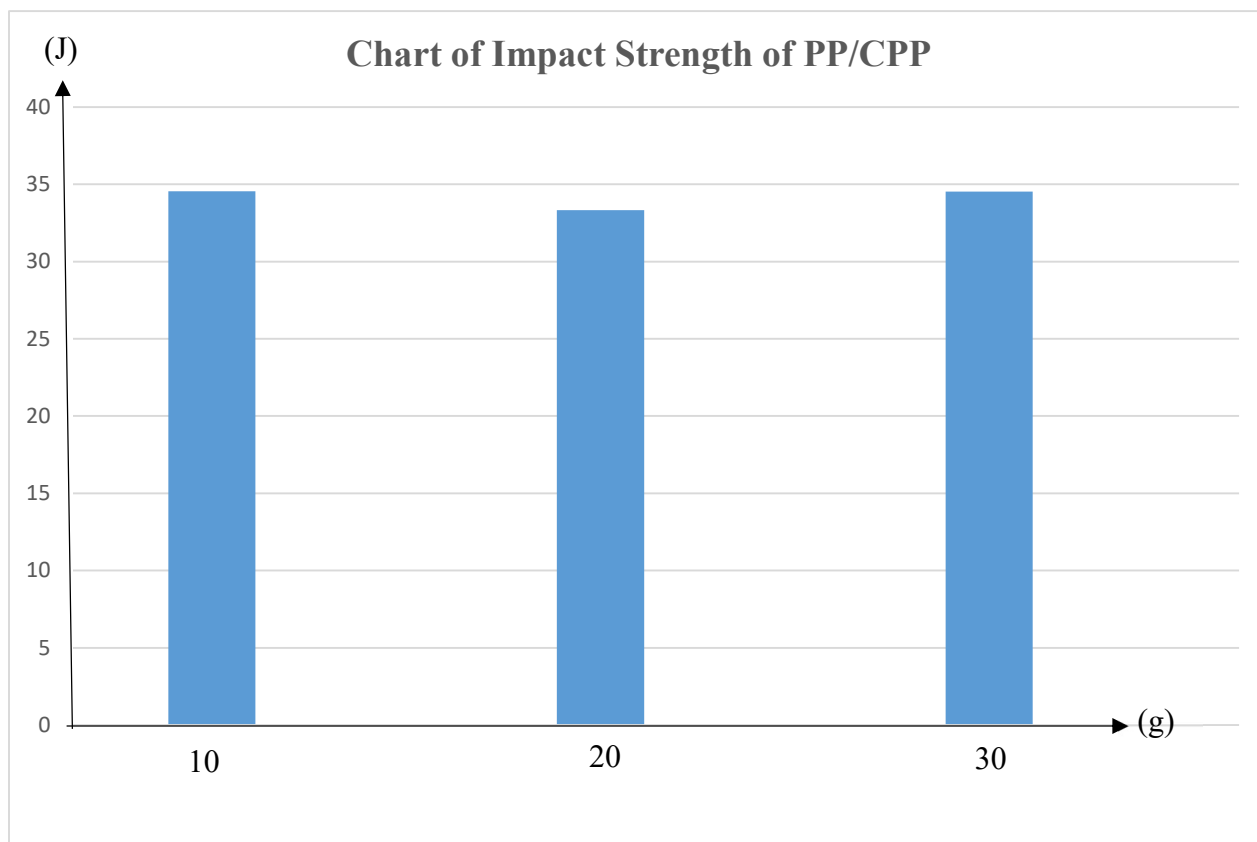


Figure 3.16: Chart Showing Impact Strength of PP/CPP Filled Composite.

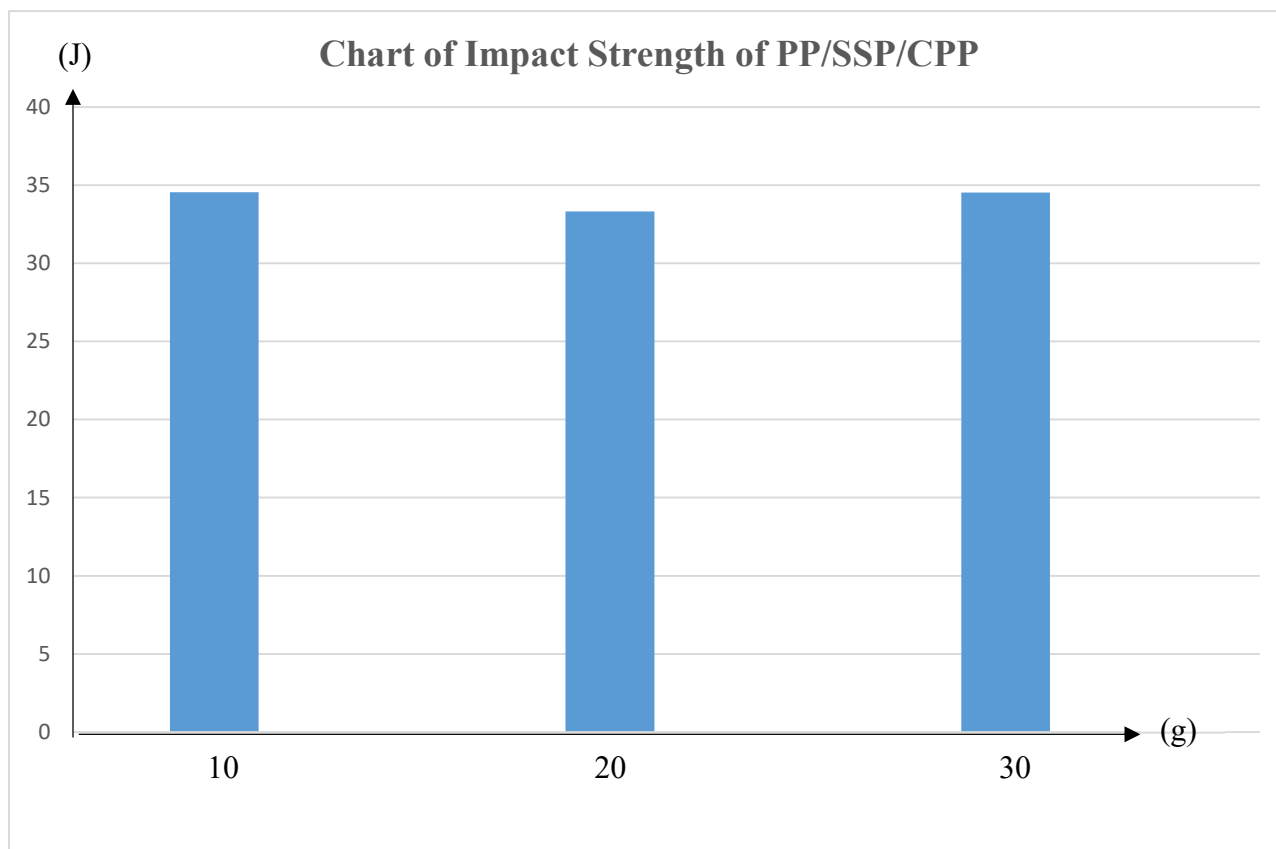


Figure 3.17: Chart Showing Impact Strength of PP/SSP/CPP Filled Composite.

3.5.1 Impact Strength

Impact strength is the amount of energy a material can withstand when a certain load is applied on it before its failure. The effect of filler loading on the impact strength of cassava peel powder (CPP), snail shell powder (SSP) and CPP/SSP filled polypropylene composites is shown in figures 3.15 to 3.17. The results showed that there was decrease in impact strength of all the composites with reference to the impact strength of pure polypropylene which is at 35.36J. Figure 3.15 Showed that the impact strength for polypropylene composite (PPC) of snail shell powder (SSP) increased with increase in filler loading at 30wt % of filler compared with 10wt %. This could be attributed to the fact that SSP has better interfacial adhesion to PP and the presence of calcium carbonate in the snail shell particles which is hard and rigid enhances the impact strength of the composites. This is similar to the study by **Lumlong et al** ,2018 However,

figure 3.16 showed the impact strength for PPC of CPP and it was observed that the impact strength decreased with increase in filler content loading; 34.00J for 10wt % of filler content to 33.86J for 30wt % of filler content. This could be attributed to the incompatibility of the filler and matrix, thus reducing the ability of the composites to withstand stress with increasing filler content loading. This is similar to the study by **lumlong et al**,(2018) and **Nwiyoronu et. al.** (2022). Also, the same observation was made on impact strength for PPC of CPP/SSP as illustrated in figure 3.17; the result showed that 160g PP/30g SSP/10g CPP was recorded to be 34.56J while 140g PP/30g SSP/30g CPP were recorded to be 34.54J. This could be attributed to poor interfacial adhesion between filler and the matrix. This is in accordance with the study by **Nwiyoronu et al**, (2022) and **Mubarak**, (2018).

Table 3.1: Result of Tensile Properties of Polypropylene and Composites

S/No	Samples/Filler content loading (wt %)	Tensile strength
1	Neat PP	35.86861
2	190g PP/10g SSP	26.56087
3	180g PP/20g SSP	21.39059
4	170g PP/30g SSP	23.80425
5	190g PP/10g CPP	19.59904

6	180g PP/20g CPP	15.35894
7	170g PP/30g CPP	15.01377
8	160g PP/30g SSP/10g CPP	21.74107
9	150g PP/30g SSP/20g CPP	22.02067
10	140g PP/30g SSP/30g CPP	23.59140

Table 3.2: Result of Flexural Properties of Polypropylene Composites

S/No	Filler content loading (wt %)	Flexural strength	Flexural modulus
1	Neat PP	161.10087	7106.62842
2	190g PP/10g SSP	33.35299	1419.35730
3	180g PP/20g SSP	29.72039	1319.90719
4	170g PP/30g SSP	18.06021	617.39955
5	190g PP/ 10g CPP	34.49079	1453.61824
6	180g PP/20g CPP	22.18974	1684.67045
7	170g PP/30g CPP	32.62745	1319.47044
8	160g PP/30g SSP/10g CPP	57.21847	2548.68374
9	150g PP/30g SSP/20g CPP	33.57255	909.02042
10	140g PP/30g SSP/30g CPP	49.93065	1915. 25307

Note; Polypropylene (PP), Cassava Peels Powder (CPP), Snail Shells Powder (SS), D (neat PP), A1 (190g PP/10g SSP), A2 (180g PP/20gSSP), A3 (170g PP/30g SSP), B1 (190g PP/10g CPP), B2 (180g PP/20g CPP), B3 (170g PP/30g CPP), C1 (160g PP/30g /10g CPP), C2 (150g PP/30g SSP/20g CPP), C3 (140g PP/30g SSP/30g CPP)

3.5.2 Tensile Strength

Tensile strength is the ability of a material to withstand maximum amount of stress while being pulled without failure. Figures 27 to 36 with

reference to Table 3.1 showed the result obtained from filler content loading of cassava peel powder (CPP), snail shell powder (SSP) and CPP/SSP polypropylene composites, it was

observed that there was a general trend of decrease in tensile strength for all the polypropylene composites when compared with the tensile strength of neat polypropylene. The tensile strength of polypropylene composites of SSP was observed to have decreased with increase in filler content loading. From the illustration in figures 28 to 30, it is clear that the higher the filler content, the lower the tensile strength. This may be attributed to the rate of dispersion and filler matrix interaction. Also, the same observation was made on the tensile strength of polypropylene composites of CPP as shown in figures 31 to 33. However, in the results obtained from tensile strength for polypropylene composites of CPP/SSP as shown in figures 34 to 36 and table 1, the tensile strength of these composites were observed to increase with increased filler content loading. 140g PP/30g SSP/30g CPP were observed to acquire higher tensile strength than others. The trend of tensile strength increased with filler content loading were related as 160g PP/30g SSP/10g CPP < 150g PP/30g SSP/20g CPP < 140g PP/30g SSP/30g CPP have. This may be attributed to better dispersion and filler matrix interaction. The results obtained similar to the study reported by Ofem, *et. al.* (2020) and Chris- Okafor, (2016).

3.5.3 Tensile Modulus

The effect of type of filler and filler content loading on the tensile modulus are shown in Table 3.1. Tensile modulus is a mechanical property that measures the resistance of material to elastic deformation. Table 3.1 showed that the tensile modulus of SPP filled composites and CPP filled composites increased as filler content is increased. The filler content is measured in 10wt %, 20wt % and 30wt % respectively. There was a slight decrease in tensile modulus strength of CPP filled of filler content at 30wt %. However, the tensile modulus strength of CPP/SSP combined filler decreased with increase in filler content loading. The results also indicated that all the filler content loading in all the polypropylene composites formulated are lower than that of the neat polypropylene except that of 160g PP/30g SSP/10g CPP that is slightly

higher. This observation may be due to the presence of crack in the matrix due to dispersion and interaction of filler particles. This is similar to the study reported by Daramola *et. al.* (2015).

3.5.4 Flexural Strength

The effect of filler loading on the flexural strength of snail shell powder (SSP), cassava peel powder (CPP) and SSP/CPP filled polypropylene composites (PPC) is shown in Table 3.2. Flexural strength is one among the useful parameters of materials in structural application which denotes the ability of material to resist deformation under load before yield point. Generally, with reference to flexural strength of neat polypropylene as presented in and Table 3.2, there was a decrease of value in the flexural strength of all the PPC. Results have shown that the flexural strength for PPC of SSP were observed to decrease with increased filler content loading. The filler content loading of all the composites is in the increasing trend of the form as 10wt % < 20wt % < 30wt %. Also, showed that the flexural strength for PPC of CCP decreased with increased filler content loading. Moreover, figure 40 showed that the flexural strength for PPC of SSP/CPP decreased as the filler content loading is increased. All these could be attributed to an uneven dispersion of the filler in the matrix of PP and non-homogeneity that led to weak bonding. This observation is similar to the study by Chai (2016) and deviated from the observation made from the study reported by Onuegbu, *et. al.* (2011) and Chris-Okafor, (2016).

3.5.5 Flexural Modulus

The flexural modulus of all the polypropylene composite decreased with reference to the flexural modulus of pure polypropylene as illustrated in Table 3.2. The flexural modulus for PPC of CPP reduced with increase in amount of filler content loading. Moreover, it was observed that the flexural modulus for PPC of SSP/CPP reduced with increase in amount of filler content loading. This is similar to the study reported by Daramola, *et. al.* (2017).

3.5.6 Elongation at Break

Deformation at break for SSP polyethylene composites is shown to rise with increased filler content in Table 3. 1, which displays the effect of filler percentage loading on the elongation at breakage properties of polypropylene composites. At the filler concentrations utilized (10g, 20g, and 30g), filler can be viewed as a structural element embedded inside the polymer matrix. However, as filler content increases, elongation at break values for CPP and CPP/SSP thermoplastic composites fall. Table 3.1 provides an example of this. There might not be enough in there to keep the polypropylene molecule around for long. Further, it is possible that a larger concentration of stain at the level of concentration under study might cause wetting between the polypropylene and the filler, resulting in essentially a non-brittle matrix. Similar findings were observed in a study by Chitra, (2016).

3.6 Conclusion.

The results obtained from the incorporation of cassava peel and snail shell powders into polypropylene matrix showed a decrease in mechanical properties of the composites. The trend was noticed in the mechanical properties such as tensile strength, tensile modulus, and elongation at break, flexural strength, flexural modulus and impact strength of all the composites. However, there was an increase in the percentage of water absorption to the duration of its immersion in water for all the composites whereas the rate of biodegradation of the composites were found to be increasing with increase in the amount of filler and the number of days it remained in the soil. This study has confirmed the effect of filler loading on the polypropylene matrix with snail shell and cassava peel powders with the characterization and monitoring test. The results summarized in Tables 3.1 and 3.2 showed the trend of reduction. The rate of biodegradation was greatly enhanced with the incorporation of filler into the matrix of polypropylene. With the exception of neat polypropylene all the prepared composites investigated showed significant water absorption with duration in water absorption test.

DECLARATION OF CONFLICTING INTERESTS.

The authors have declared no conflict of interests.

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